

# Blocco 3

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## Esercizio 1.

*Proof.*

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**Warning.** In the following we will adopt the following conventions:

- In each exercise,  $P$  will denote the poset used
- When something coincide when seen in  $V$  or in  $V[G]$  we we'll not specify where we are looking at it. So for example we we'll always write  $\omega$  instead of  $\omega^V, \omega^{V[G]}$ .

For the next problem you can use that for inaccessible  $\lambda$ ,  $Col(\omega_1, < \lambda)$  has the  $\lambda - cc$  and if  $G$  is a  $Col(\omega_1, < \lambda)$ -generic filter over the ground model  $V$ , then  $V[G] \models \lambda = \omega_2$ .

**Esercizio 2.** Suppose that  $\lambda$  inaccessible. Show that  $Col(\omega_1, < \lambda)$  forces that CH holds.

*Proof.* Since  $\omega < \lambda$ , we have by inaccessibility of  $\lambda$  that  $|2^\omega| < \lambda$ . Let  $G$  be a generic filter for  $P$ . First we claim that, by  $\omega_1$ -closure of  $P$ , we have  $(2^\omega)^V = (2^\omega)^{V[G]}$ .

- $(2^\omega)^V \subseteq (2^\omega)^{V[G]}$ : because  $V \subseteq V[G]$ .
- $(2^\omega)^V \supseteq (2^\omega)^{V[G]}$ : Let  $V[G] \models A \subseteq \omega$ . We want to show that  $A \in V$ . Let  $\dot{A}$  be a  $P$ -name over  $V$  for  $A$ . Now consider  $\langle p_n : n < \omega \rangle$ , defined inductively as follows:

1.  $n = 0$ :

- if  $V[G] \models 0 \in A$  take  $p_0 \in G$  such that  $p_0 \Vdash \check{0} \in \dot{A}$
- if  $V[G] \models 0 \notin A$  take  $p_0 \in G$  such that  $p_0 \Vdash \check{0} \notin \dot{A}$

2.  $n \Rightarrow n + 1$ :

- if  $V[G] \models n + 1 \in A$  take  $p_{n+1} \in G, p_{n+1} \leq p_n$  such that  $p_{n+1} \Vdash n \check{+} 1 \in \dot{A}$
- if  $V[G] \models n + 1 \notin A$  take  $p_{n+1} \in G, p_{n+1} \leq p_n$  such that  $p_{n+1} \Vdash n \check{+} 1 \notin \dot{A}$

Now let  $\bar{p} \in G, \bar{p} \leq p_n$  for all  $n < \omega$ . Such an element exists because one can consider  $p' = \bigcup_n p_n, D = \{q : q \leq p'\} \cup \bigcup_n \{q : q \perp p_n\}$ . We claim that  $D$  is dense. In fact, let  $r \in P$ . If  $r \perp p_n$  for some  $n < \omega$  we already have  $r \in D$ . So we can assume  $r \perp p_n$  for all  $n < \omega$ . Define  $\langle r_n : n < \omega \rangle$  as follows:

1.  $n = 0$ : take  $r_0 \leq r, p_0$

2.  $n \Rightarrow n + 1$ :

- if  $r_n \perp p_{n+1}$  then we have  $r_n \in D, r_n \leq r$  so one can stop.
- othewise take  $r_{n+1} \leq r_n, p_{n+1}$ .

If the sequence stopped we proved the density, so assume the sequence did not stop. In this case take  $r' = \bigcup_n r_n$ . Then we have  $r' \leq p_n$  for all  $n < \omega$ , which implies  $r' \leq p'$ . So we have  $r' \in D, r' \leq p_n$  for all  $n < \omega$ , which proves the density of  $D$ .

Since  $D \in M$  is dense and  $G$  is generic we have  $G \cap D \neq \emptyset$ . Let  $\bar{p} \in D \cap G$ . Since  $p_n \in G$  for all  $n < \omega$  it must be  $\bar{p} \leq p_n$  for all  $n < \omega$ . So  $\bar{p}$  is the element we were looking for.

Now consider  $B = \{n : \bar{p} \Vdash \check{n} \in \dot{A}\}$ . Note that  $B \in M$  since one can use the equivalent internal definition of forcing. We claim  $B = A$ .

- $A \subseteq B$ :  $n \in A \Rightarrow \exists r \in G(r \Vdash \check{n} \in \dot{A}) \Rightarrow \bar{p} \Vdash \check{n} \in \dot{A}$  because otherwise  $\bar{p}, r \in G$  would be incompatible.
- $A \supseteq B$ :  $n \in B \Rightarrow \bar{p} \Vdash \check{n} \in \dot{A} \Rightarrow V[G] \models n \in A$ .

This means that  $\mathcal{P}(\omega)^V = \mathcal{P}(\omega)^{V[G]}$ . Now observe that  $|2^\omega|^V \geq |2^\omega|^{V[G]}$  and since  $|2^\omega|^V \in \lambda$  we get  $|2^\omega|^{V[G]} \in \lambda$ . Furthermore from  $V[G] \models \lambda = \omega_2$  we get  $V[G] \models |2^\omega| < \omega_2$ . Since  $V \models ZFC$  and  $2^\omega \geq \omega_1$  is a theorem of ZFC, we also get  $V[G] \models |2^\omega| \geq \omega_1$ . So we conclude  $V[G] \models |2^\omega| = \omega_1$ .

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